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Great Observatories: The Spitzer Space Telescope

The Infrared Explorer

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Introduction

In August 2003, a rocket rose from Florida carrying a telescope that would never orbit the Earth. The Spitzer Space Telescope, the infrared member of NASA's Great Observatories, was sent instead into a heliocentric orbit, trailing the Earth around the Sun, so that it could cool itself away from the warm planet's glare and watch the universe in the light of heat. For sixteen and a half years, until January 2020, it did exactly that, and the results reshaped almost every corner of astronomy: the first light ever measured from planets around other stars, the seven Earth-sized worlds of TRAPPIST-1, the carbon spheres called buckyballs floating between the stars, Saturn's enormous hidden dust ring, and the hidden structure of the Milky Way itself, seen at last through the dust that blinds ordinary telescopes.

This volume tells the observatory's story: its origin in the long struggle to do astronomy above the atmosphere; its namesake, Lyman Spitzer, the astrophysicist who proposed space telescopes in 1946, three decades before anyone could build one; the engineering that made it the first telescope launched warm and cooled in space, saving its coolant for the science; the infrared physics that explains what heat-light can reveal; and the discoveries, gathered here from the mission's sixteen-year run, that made a modestly sized telescope one of the most productive scientific instruments ever flown.

The chapters follow the series template: the discovery, the understanding, the characteristics, the science, the observation methods, the important features and discoveries, the context among the Great Observatories, the modern understanding, the research programs, the challenges and mysteries, the future and the successors, the trivia, and the significance. What binds them together is the lesson the mission's whole history teaches: the cold universe is a different universe, and the telescope that learned to live in the cold opened it.

Chapter 1: The Discovery of The Spitzer Space Telescope

The discovery of the Spitzer Space Telescope, like that of most observatories, is the story of an idea's long journey to a launch pad, and the idea was older than the space age itself.

The founding document came from the mission's own namesake. In 1946, Lyman Spitzer, a young astrophysicist already known for his work on interstellar matter, wrote a paper arguing for the astronomical advantages of an extraterrestrial observatory: above the atmosphere, a telescope escapes the blurring of seeing, the absorption of the ultraviolet and infrared, and the background glow of the sky, and could answer questions no ground-based instrument could touch. The man himself deserves the paragraph the namesake owes. Lyman Spitzer, born in 1914, was a theorist of the interstellar medium who became Princeton's astrophysics chairman at thirty-one, spending the war years on underwater sound research before returning to the stars; his postwar career ran two great lines at once, the space telescope advocacy that began with the 1946 paper, and the plasma physics that produced the stellarator, the magnetic bottle for fusion whose descendants carry his design's name today. The double career was one conviction in two laboratories: hot matter, whether in the stars or in a fusion machine, was the physics worth bottling, and the space telescope was the same program's instrument, the atmosphere being the bottle whose cork the astronomer wanted drawn. He died in March 1997, months after Hubble's second servicing mission restored its vision; the naming contest that gave his name to the infrared observatory came two years later, and the mission's scientists noted the fit: the infrared telescope, the field Spitzer's interstellar work had helped define, would carry the name of the man who made space telescopes credible.
Spitzer spent the following half century championing the idea, most famously as the intellectual father of the Hubble Space Telescope, this series' own volume tells that story, and he lived long enough to see Hubble's first repair mission succeed; he died in 1997, and when NASA renamed the Space Infrared Telescope Facility in his honor, the project's engineers said the tribute was owed to the man who had started it all.

The infrared road to Spitzer ran through a chain of earlier missions. Infrared astronomy from the ground is nearly hopeless for faint sources: the atmosphere's own warmth, and the telescope's, swamp the detectors, and the water vapor blocks most of the wavelength range; the science therefore took to balloons, aircraft, and small satellites through the 1960s and 1970s, each flight proving the sky's infrared riches. The breakthrough came with the InfraRed Astronomical Satellite, IRAS, launched in 1983, a joint Dutch-British-American cryogenic survey that mapped the whole sky in infrared light and found it crowded: hundreds of thousands of sources, star-forming regions blazing through the dust, galaxies glowing in hidden warmth, and the first Vega-like debris disks, the discovery that other stars have their own material outskirts. IRAS's own story deserves a line, the survey that made the case. Built in a three-nation partnership and launched in January 1983, the satellite carried a small

helium-chilled telescope and, in ten months of cryogenic life, mapped the whole sky at four infrared wavelengths; the catalogues ran to hundreds of thousands of sources, and the mission's atlas, the first infrared sky, showed the galaxy as a blaze of dust-warmed light with the star-forming regions, the plane's hidden clouds, the bright nuclei of galaxies all visible for the first time as a population. The survey's list of firsts, the Vega dust disk above all, gave the field its headline science, and the era's planners read the lesson plainly: the infrared sky was rich, and the follow-up observatory would be worth any amount of engineering.
IRAS proved the field; Europe's Infrared Space Observatory, ISO, launched in 1995, proved the follow-up, observing chosen targets with spectrographs and imaging through the mid-infrared; and NASA's next step, the Space Infrared Telescope Facility, SIRTF, was approved and designed as the cryogenic flagship the field wanted.

SIRTF's own history is a study in scientific resilience. The mission as first conceived in the 1980s was a large, Shuttle-launched, helium-heavy observatory, and its cost estimates climbed until, in 1991, the program was reshaped entirely: the telescope would be smaller, launched on a Delta II rocket, and, in the design's masterstroke, launched warm and cooled in the vacuum of space, so that most of the liquid helium needed to chill the instruments would not be spent cooling the structure on the ground. The redesign's orbit was the other masterstroke: instead of circling the Earth, where the planet's warmth and the day-night cycling would limit the mission's life, the telescope would be sent into a heliocentric orbit, drifting away from Earth on a trailing path, an orbit no observatory had used before, allowing stable, continuous cold operations for years.

The redesign's politics deserve one more paragraph, the 1991 decision's context. The program's downsizing, forced by the era's budget crisis and the Shuttle fleet's uncertain future, was contested and painful: the infrared community had dreamed the large observatory for a decade, and the smaller vehicle meant a smaller mirror, fewer instruments, and a survey-and-pointed balance different from the original; the decision's defense, made by the program's scientists and accepted by the community, was the mission-philosophy's: a smaller telescope in the right orbit, cold and stable, would outperform the bigger plan's promises, and the mission's subsequent history, the sixteen years and the founding discoveries, was the defense's vindication. The lesson entered the field's planning culture, the mid-scale missions' era citing it, and the infrared community's own telling of the story calls the 1991 decision the field's luckiest loss.
The launch came on August 25, 2003, from Cape Canaveral, and the commissioning went by the book: the telescope cooled, the instruments verified, and in December 1999, before the hardware was even launched, the spacecraft had received its final name, chosen through a public naming contest in honor of Lyman Spitzer. The mission as flown had a planned cryogenic life of two and a half years, with the hope of five; the helium lasted until May 2009, nearly six years, and the warm mission that followed, running the camera's two shortest-wavelength channels without coolant, extended the observatory's life to more than sixteen years in total. The discovery of the Spitzer Space Telescope, in the sense that matters, is thus the whole arc: an idea in 1946, a proof in 1983, a redesign in 1991, a launch in 2003, and a mission that outlived its plans by a decade, the subject of the chapters that follow.

A paragraph on the long fuse, because the observatory was proposed before the machines that could carry it existed. The argument for an observatory above the atmosphere was written in the 1940s, years before the first satellite had flown at all, a physicist's paper making the case that a telescope in space would see, free of the air's blur and the air's blocking, what no ground-based instrument could; the idea then waited through decades of rocketry, budgets, and competing designs before its infrared descendant was finally built and flung into orbit. The discovery's note for the file: this machine was old at birth, its intellectual blueprint drawn a half-century before its hardware, and the naming of it for the man who wrote that early argument was the field's way of admitting that the patience of an idea had outlasted everyone's impatience with it.